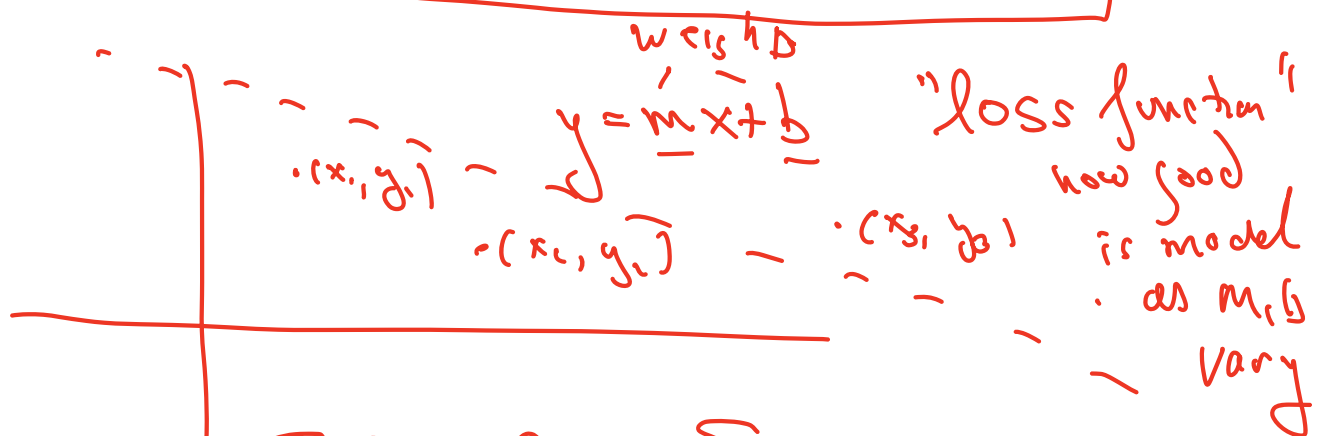


From last time:

$$\begin{aligned} m S_{xx} + b S_x &= S_{xy} \\ m S_x + N b &= S_y \end{aligned}$$



$$\begin{aligned} S_{xx} &= \sum x_i^2 & S_{xy} &= \sum x_i y_i \\ S_x &= \sum x_i & S_y &= \sum y_i \end{aligned} \quad N = \# \text{ pts}$$

$$m = \frac{S_y - Nb}{S_x}$$

$$\frac{S_y - Nb}{S_x} \cdot S_{xx} + b S_x = S_{xy}$$

$$\frac{S_y S_{xx}}{S_x} - \frac{N S_{xx}}{S_x} b + b S_x = S_{xy}$$

$$b \left(S_x - \frac{N S_{xx}}{S_x} \right) + \frac{S_y S_{xx}}{S_x} = S_{xy}$$

$$b = \frac{S_{xy} - S_y S_{xx} / S_x}{S_x - N S_{xx} / S_x}$$

$$b = \frac{S_x S_{xy} - S_y S_{xx}}{S_x^2 - N S_{xx}}$$

$$S_x^2 = N S_{xx} \quad ? \quad 1, 2, 3$$

$$(\sum x_i)^2 = N \sum x_i^2$$

~~6~~

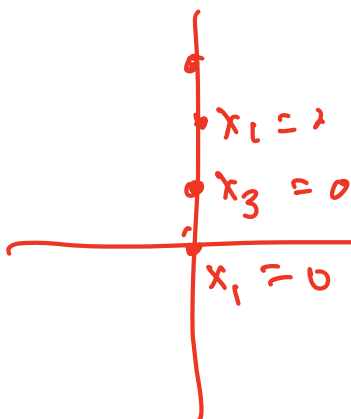
$$C = x_1 = x_2 = \dots = x_N$$

$$\sum x_i = N C$$

$$\sum x_i^2 = N C^2$$

$$(\sum x_i)^2 = N^2 C^2$$

$$N \sum x_i^2 = N^2 C^2$$



$$\{(x_i, y_i)\}_{i=1}^N \quad \text{data}$$

$$y = mx + b$$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} \quad M = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_N & 1 \end{bmatrix}$$

$N \times 1$ $N \times 2$

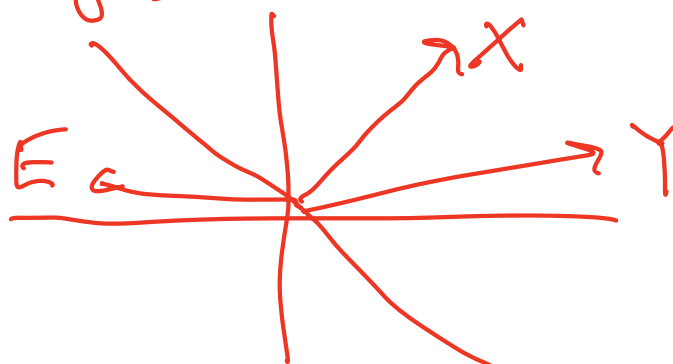
$$\hat{Y}_{m,b} = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_N & 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = W$$

$\begin{bmatrix} m \\ b \end{bmatrix}$

$$\begin{bmatrix} m \\ b \end{bmatrix}$$

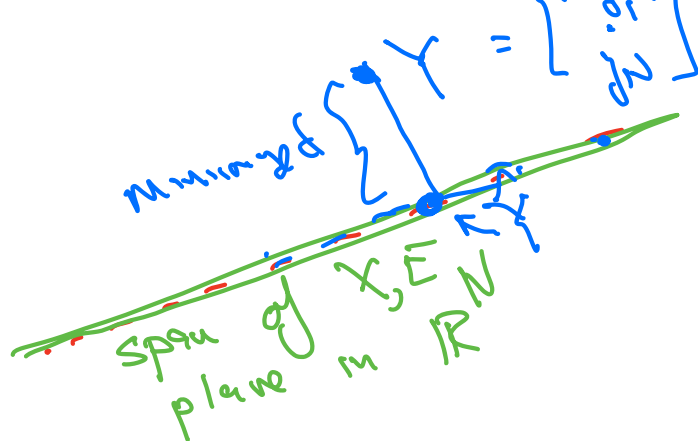
$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$$

$$E = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$



$$Y = mX + bE$$

If x_i, y_i well on $y = mx + b$



side view

Find "best" m, b so that
 $\hat{Y} = mX + bE$ is "best"!

$$\underbrace{\sum (\hat{y}_i - y_i)^2}_{\sim m, b} \quad \text{minimized}$$

$$Y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \quad M = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_N & 1 \end{bmatrix}$$

$$W = \begin{bmatrix} m \\ b \end{bmatrix}$$

Solve:

Find W so that

$$\|Y - MW\|^2$$

is minimized.

$$\left\| \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \right\|^2 = \sum a_i^2$$

Multivariate Linear Regression

single target y features
 several input variables $x_1, x_2, \dots, x_k$

$$y = m_1 x_1 + m_2 x_2 + \dots + m_k x_k + b$$

target ← weights ← bias

Data $(y_1, a_{11}, a_{12}, \dots, a_{1k})$
 $(y_2, a_{21}, a_{22}, \dots, a_{2k})$

data matrix X
 k columns
 N rows

$$\begin{bmatrix} x_{11} & x_{12} & \dots & x_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \dots & x_{Nk} \end{bmatrix}$$

N points in k dim'l space
 k pts in N dim'l space

extended X^*

$$\begin{bmatrix} x_{11} & x_{12} & \dots & x_{1k} & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{N1} & x_{N2} & \dots & x_{Nk} & 1 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ \vdots \\ b \end{bmatrix} = Y$$

$\hat{Y}$ predicted

$Y = X^* W$

Goal: "best" choice of W .

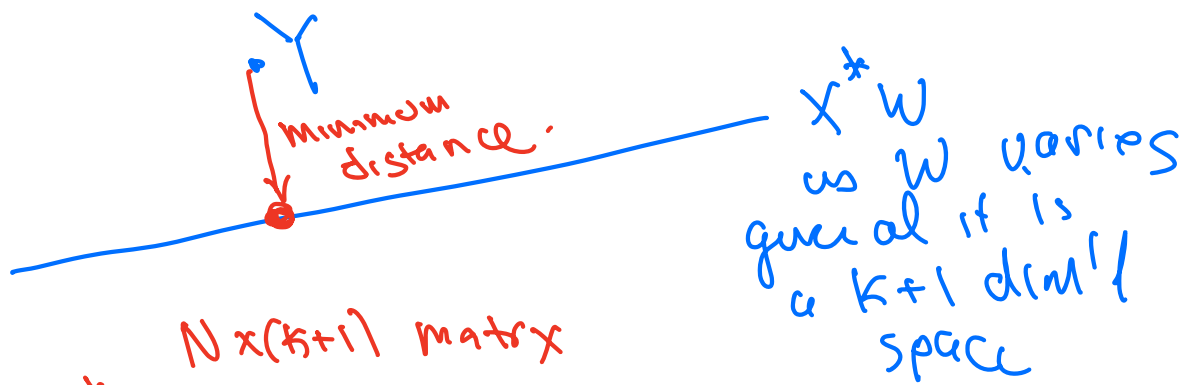
$$MSE = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$= \frac{1}{N} \sum_{i=1}^N (y_i - (m_1 x_{i1} + m_2 x_{i2} + \dots + b))^2$$

minimize MSE.

$$\text{MSE} = \frac{1}{N} \|Y - \hat{Y}\|^2$$

$$= \frac{1}{N} \|Y - X^* W\|^2$$



$N \times (k+1)$ matrix

$$X^* = (x_{ij}^*) \quad W_j \quad (k+1) \times 1 \text{ matrix}$$

$$(X^* W)_j = \sum_{s=1}^{k+1} x_{js}^* w_s$$

$$A = (a_{ij}) \quad N \times k$$

$$B = (b_{jk}) \quad k \times t$$

$$(AB)_{rk} = \sum_{j=1}^k a_{rj} b_{jk} \quad \begin{pmatrix} a_{r1} & a_{r2} & \dots \end{pmatrix} \begin{pmatrix} b_{1r} \\ b_{2r} \\ \vdots \end{pmatrix}$$

$N \times t$

$$(Y - X^* W)_j$$

$$= \left(y_j - \sum_{s=1}^{k+1} x_{js}^* w_s \right)$$

$$L = \text{MSE} = \frac{1}{N} \sum_{j=1}^N \left(y_j - \sum_{s=1}^{k+1} x_{js}^* w_s \right)^2$$

$$\frac{\partial L}{\partial w_t}$$

$$t = 1, \dots, k+1$$

e.g. $f(x, y, z) \rightsquigarrow \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{bmatrix}$